

Sample Project Documentation

AI StudyBuddy: AI-Powered Study Assistant for Smarter Learning

Project Description:

AI StudyBuddy is a backend REST API powered by Generative AI that automates study preparation for students. The platform allows authenticated users to upload study materials (.txt, .md, .pdf) and leverages Google Gemini 2.5 Flash API to generate four types of AI-powered learning resources: concise summaries, Q&A flashcards, multiple-choice quizzes, and personalized study plans. Built with Node.js, Express.js, and MongoDB Atlas, the API features JWT-based authentication with access and refresh tokens, Role-Based Access Control (student/admin), and persistent storage of all AI-generated content. With 18 REST API endpoints across authentication, material management, AI generation, and admin modules, AI StudyBuddy provides a comprehensive backend solution for intelligent learning support.

Scenarios:

Scenario 1: A student uploads their biology notes as a .txt file and requests an AI summary. The system sends the content to Google Gemini, which returns a concise bullet-point summary. The student then generates 5 flashcards for quick revision — the AI returns a JSON array of question-answer pairs covering all key concepts from the material.

Scenario 2: A student preparing for exams generates a 5-question MCQ quiz from their uploaded chemistry notes. The AI returns multiple-choice questions with 4 options each and the correct answer highlighted. The student uses these to self-assess their understanding before the exam.

Scenario 3: A student with 7 days until their exam and 2 hours of study time per day requests a personalized study plan. The AI analyzes their uploaded material and generates a structured day-by-day schedule with specific topics, revision activities, and practice sessions tailored to their constraints.

Exploring the API Endpoints:

Authentication Endpoints:

- POST /api/auth/register — Register with name, email, password. Returns JWT tokens.
- POST /api/auth/login — Login with email and password. Returns JWT tokens.
- POST /api/auth/refresh — Refresh expired access token using refresh token.
- POST /api/auth/logout — Stateless logout.

Material Management Endpoints (requires JWT):

- POST /api/materials/upload — Upload study material (.txt/.md/.pdf, max 5MB).
- GET /api/materials — List all user's materials.
- GET /api/materials/:id — Get full material with all AI-generated content.
- DELETE /api/materials/:id — Delete a material.

AI Generation Endpoints (requires JWT):

- POST /api/materials/:id/summarize — Generate AI summary.
- POST /api/materials/:id/flashcards — Generate AI flashcards (JSON).
- POST /api/materials/:id/quiz — Generate AI MCQ quiz (JSON).
- POST /api/materials/:id/study-plan — Generate personalized study plan.
- POST /api/ai/flashcards — Standalone flashcards (materialId in body).
- POST /api/ai/quiz — Standalone quiz (materialId in body).
- POST /api/ai/study-plan — Standalone study plan (materialId in body).

Admin Endpoints (requires admin role):

- GET /api/admin/users — List all users.
- DELETE /api/admin/users/:id — Delete user and their materials.
- GET /api/admin/stats — Total users and materials count.

Conclusion:

AI StudyBuddy is a comprehensive backend API that demonstrates the practical application of Generative AI in education. By combining Node.js, Express.js, MongoDB, and Google Gemini AI, the project automates the most time-consuming aspects of study preparation — summarization, flashcard creation, quiz generation, and study planning. The modular MVC architecture, secure JWT authentication, and clean API design make the project maintainable and extensible. Future enhancements include a React/Next.js frontend, AI chatbot for doubt solving, progress tracking, multi-language support, and cloud deployment for production use.